

Immersive Telerobotics with Haptic Feedback

ENG4701 Project Proposal

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Acknowledgement of Country

We would like to acknowledge the traditional custodians of the Bunurong people of the Kulin Nations, on whose ancestral lands we convene for our final year project. We offer our profound respect and gratitude to the Ancestors and Elders — past, present and emerging — who have stewarded the Bunurong Country for countless millennia.

As one of the world's oldest civilisations, we recognise the Aboriginal and Torres Strait Islander peoples as the first engineers, scientists, innovators and inventors of this nation. Throughout the ages, the Ancestors and Elders developed tools to help make everyday life easier. One such tool is a spear thrower, known as a woomera or miru [seen in Figure 1 and Figure 2], a device developed over 10,000 years ago to help hunters throw a spear much further than normally possible. This allowed for greater force, speed (up to 150 km/hr), and distance when launching the spear, especially through water [1][2]. Its unique and multipurpose design enables it to be a versatile tool, also being able to be used as a weapon or a shield on its own. As we progress with our final year project, we acknowledge our historical and ongoing responsibility to create useful tools that further ease the lives of our community, just like the traditional custodians of this land.



Figure 1: A range of different styles and types of woomeras. [1].



Figure 2: A hunter throwing a three-pronged spear using a woomera, 1983. [2].

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1 Introduction

The human hand is an exceptionally sensitive instrument, with 17,000 touch receptors and free nerve endings in each palm alone [3], allowing us to discern between pressure, movement and vibration, to list a few. Touch, the first of the five senses developed during gestation [4], plays a crucial role in relaying a plethora of multidimensional information to the brain to inform decision-making processes and facilitates fine motor control.

In the context of teleoperation — a sector valued at US \$494.7 billion in 2023 and projected to grow by approximately 52% over the next eight years [5] — the role of tactile feedback becomes increasingly critical. Teleoperation serves as a tool to remotely operate equipment and machinery, a capability that is particularly advantageous in training and daily operations in high-risk environments such as hazardous material handling in laboratories and nuclear facilities. Imagine an operator tasked with controlling such equipment based solely on visual input. The absence of tactile feedback—a sense we rely on for both simple and complex tasks—can lead to increased cognitive load, diminished operational accuracy and significant operational risks, highlighting the significant need for immersive sensory integration.

As teleoperation becomes increasingly integral to industries from remote work to advanced training simulations, the need for effective, affordable solutions that enhance sensory feedback is paramount. Recent advancements in virtual reality (VR) and haptic technology have introduced promising solutions aimed at augmenting sensory feedback in teleoperation but many existing products come with high costs or lack the immersive qualities required for optimal performance. Addressing these limitations, this project proposes the development of an affordable haptic glove that integrates both tactile and pseudo-haptic feedback mechanisms with commercially available VR systems. The overall aim of the project is to develop an integrated virtual reality system that enhances user control and precision in teleoperating a robotic arm through real-time tracking and manipulation.

The system developed will utilise and be implemented for use in the Digital Twin Lab of the Monash Smart Manufacturing Hub (MSMH) which is home to four KUKA iiwa robotic arms. This project system developed will allow Monash staff and students to remotely control a two-finger end-effector KUKA robotic arm, enabling operations such as grabbing, picking up, moving, and pushing objects, all with enhanced haptic and pseudo-haptic feedback. Currently, these tasks are managed using a cognitively demanding joystick or iPad. By advancing the integration of haptic feedback in teleoperation, this project aims to contribute to the development of more intuitive, affordable and reliable remote operation systems, setting the stage for future advancements in haptic technology.

2 Aims and Objectives

2.1 Aims

The overall aim of this project is to develop an integrated virtual reality system that enhances user control and precision in teleoperating a robotic arm through real-time tracking and manipulation. The system will enable users to teleoperate robotic arms more intuitively due to combined immersive features, with improved feedback mechanisms that reduce cognitive strain. Specifically, the project aims:

1. To control a robotic arm and end-effector in real-time using a VR headset and hand-tracking.
2. To accurately determine and realistically simulate the sensations experienced by the robotic arm's end effectors.
3. To provide operators with real-time sensory feedback to aid in precise manipulation and decision-making.
4. To evaluate by quantifying how the immersive features developed ease the cognitive strain of the operator, compared to a baseline of no immersive features.

Achieving these will establish the foundations for future maintenance, enhancements and combination of more immersive features.

2.2 Objectives

To manage the project's scope effectively, it is divided into the following main areas:

1. Integration of Robotic Arm Control via VR Headset
2. Development of a Haptic Glove
3. Implementation of a Haptic Suit
4. Integration of Haptic Glove and Suit with VR headset and Robotic Arm
5. Evaluating the Benefits of Immersive Feedback on Operators
6. Stretch-Goal - Mind-Controlled Robotic Arm Operation

The objectives of each area are explored in Section 2.2.1 to Section 2.2.6.

2.2.1 Integration of Robotic Arm Control via VR Headset

A VR headset will be used to provide pseudo-haptic feedback and enable teleoperation through real-time hand-tracking. In order for this to occur, the following objectives need to be achieved:

1. To understand the robotic arm and VR headset hardware by investigating the functionality (i.e. sensors available) and limitations of the robotic arm (Kuka LBR iiwa), end effectors (Robotiq 2F-85 & 2F-140), the VR headsets (Meta Quest 3), completing this research within one week.
2. To learn Unity and how to integrate Oculus VR with Unity within one week.
3. To create a digital twin of the robotic arm for the VR headset using the Zed-2i stereo cameras within one week.
4. To learn to control the robotic arm through the VR headset using Zed-2i stereo cameras within one week.
5. To investigate the use cases, advantages, disadvantages, costs, and implementation feasibility of three pseudo-haptic features that could be incorporated into the robotic arm and VR headset within three days.
6. To procure the necessary pseudo-haptic hardware, and implement at least two pseudo-haptic features within sixty-three days (allowing enough time for procurement).
7. To test and validate the pseudo-haptic features to ensure effective robotic arm control via the VR headset, within one week.

2.2.2 Development of a Haptic Glove

A two-finger haptic glove will need to be developed to provide users with real-time feedback through touch. This will help users understand when the robotic end effectors interact with objects, including the force and restriction applied. To achieve this, the following objectives must be met:

1. To understand the different technologies available by conducting a comprehensive literature review of at least five different haptic feedback technologies, documenting their use cases, advantages, disadvantages, costs, and implementation challenges, to be completed within one week.
2. To understand the commercially available haptic gloves by conducting a state of the field review of at least three commercially available haptic gloves, focusing on their feedback technology, costs, and effectiveness, and create a comparison matrix, within one week.
3. To create parts and assembly models using a 3D Computer-Aided Design (CAD) tool, SolidWorks, for a prototype of the two-finger haptic glove, incorporating at least two various types of haptic feedback investigated, with the CAD design to be completed within two weeks.
4. To source all required components for the glove, determining which can be manufactured in-house and which need to be procured, and finalise sourcing within sixty-three days (allowing enough time for procurement).
5. To assemble the glove components and integrate them with software and electrical systems, ensuring functionality through testing, to be completed within two weeks.

2.2.3 Implementation of a Haptic Suit

A commercially available haptic suit, Teslasuit, will be used in this project to provide real-time feedback on the robotic arm's motion constraints, enhancing user awareness of the arm's limitations. To achieve this, the following objectives must be met:

1. To investigate the functionality, use cases, limitations, and risks of available haptic suits and obtain necessary training to operate the suit, with a detailed report and training completion within one week.
2. To understand the development documentation and write code to manage haptic feedback for the arm portion of the suit, ensuring proper control of features within one week.
3. Test and validate the suit's feedback accuracy to ensure it effectively enhances user awareness of the robotic arm's motion constraints, with comprehensive testing completed within two weeks.

2.2.4 Integration of Haptic Glove and Suit with VR headset and Robotic Arm

Once the VR headset (Section 2.2.1), haptic glove (Section 2.2.2) and suit (Section 2.2.3) objectives have been achieved, they will need to be integrated together to be able to be used to control the robotic arm. As a result, these objectives must be met:

1. To develop middleware to enable communication between the haptic glove and Unity via Raspberry Pi using OPC Unified Architecture, with middleware functionality established within one week.
2. To synchronise the haptic glove with the haptic suit and VR headset to ensure seamless interaction between all of devices, completing the integration within one week.
3. To investigate and test optimal methods for translating robotic arm data into haptic responses, with findings documented within one week.
4. To test and validate the overall integration of the haptic feedback from the glove and suit with the VR headset control and pseudo-haptic feedback mechanisms to ensure smooth and effective operation, within one week.

2.2.5 Evaluating the Benefits of Immersive Feedback on Operators

Once the system is fully functioning after achieving the objectives in Section 2.2.4, then a study can be done on the effectiveness of the additional immersive features added in the manipulation and control of the robotic arm. To successfully achieve this, the following objectives must be met:

1. To determine and define the target user (i.e. the operator) that would use the system and develop a sample size pool of operators by advertising and interviewing a range of candidates, within nine days.
2. To develop a list of quantitative and qualitative data that needs to be collected and the manner of which it will be collected (i.e. test plan), within two days.
3. To perform the baseline test of just using the VR headset to control the robotic arm with the available sample size within two weeks.
4. To perform the main test of incorporating the haptic and pseudo-haptic features to control the robotic arm with the available sample size within two weeks (simultaneously while doing Objective 3 in Section 2.2.5).
5. Run statistical modelling on the data to determine whether any conclusions could be drawn, within one week.

2.2.6 Stretch-Goal - Mind-Controlled Robotic Arm Operation

If time permits, a stretch goal is to explore using brain signals to control robotic movement. This could contribute to the overall aim of this research problem as it could enhance the system's overall functionality. To achieve this, the following objectives must be met:

1. To research existing VR mind-control solutions, focusing on their functionality, development processes, and costs, with a comprehensive report completed one week.
2. To procure the necessary hardware for training a neural network, and collect operator data on simple tasks with the robotic arm within two weeks.
3. To train a neural network using the collected data to control the robotic arm, implement the model, and conduct testing and validation within one week.
4. Evaluate the effectiveness of mind-control in reducing cognitive strain and its overall feasibility by following a similar procedure as outlined in Section 2.2.5, within thirty-five weeks.

3 State of the Field

Over the past few decades, teleoperation systems have become increasingly prevalent. In particular, robot-assisted teleoperation systems have seen significant growth, largely due to their enhanced accessibility and ability to perform tasks in inaccessible or hazardous environments, amongst other reasons. These systems have found applications across various sectors, including:

- Exploration: Teleoperated robots are used in deep space, aerial, and underwater missions, enabling exploration in extreme and remote environments. For example, the underwater teleoperated robots, Remotely Operated Vehicles (ROVs), that have been significant in deep-sea exploration [6].
- Healthcare: The healthcare sector has greatly benefited from teleoperated robotic systems, performing over 700,000 precision surgeries [7] annually with technologies like the da Vinci Surgical System [8]. These robotic systems have also been explored in telemedicine by enabling remote patient examinations, particularly crucial during the COVID-19 pandemic for reducing direct contact and maintaining safety [9]. Their use is also being considered in rehabilitation using additional feedback [10].
- Infrastructure: They are used for inspections in hazardous environments in live-line maintenance in the power industry [11], pipeline inspection in the oil and gas sector, and tasks within nuclear facilities [12], where they ensure safety and enhance operational efficiency through advanced autonomy and sensing capabilities.
- Disaster Response: Search and rescue operations increasingly rely on teleoperated robots to access dangerous areas [13].
- Education: Remote laboratories and virtual classrooms have the potential of utilising teleoperation for hands-on learning experiences [14].
- Assistive Technology: Support for disabled and elderly individuals is enhanced through teleoperated robotic aids, improving independence and quality of life. This is achieved by the use of socially assistive [15] and telepresence robots.

For such teleoperated activities, some key objectives emerge while developing these teleoperated systems. These include ensuring (i) robustness, (ii) sense of presence, (iii) task performance and (iv) operational efficiency [16], with a sense of presence being the most critical.

Regarding a sense of presence, integrating visual input, like a Head Mounted Display (HMD), can help increase immersiveness and the sense of presence of the user at the remote location. Ensuring a sense of presence leads to improved task performance. This is marked by various metrics such as reduced error rate, time of completion and the efficiency with which tasks are performed. Ensuring these objectives are met foster a more intuitive and responsive interaction between the user and the remote environment. In tasks involving high Situation Awareness such as the control of a telerobotic system in hazardous environments or during complex surgical procedures, enhancing the sense of presence and responsiveness becomes a crucial factor. As such, this leads to improved efficiency by reducing reaction times and better decision-making which meets the first objective of robustness. As a result, safety of systems are addressed along with decreased operational risks.

Another way of adding immersiveness is through the use of incorporation of haptic feedback. This includes haptic feedback such as cutaneous (tactile) feedback for the hands and kinesthetic feedback for capturing movements and forces. Electrical Muscle Stimulation (EMS) feedback on the arms to simulate touch sensations and muscle engagement adds another layer of realism to the experience. These different types of haptic and EMS feedback are further discussed in Section 3.2 and Section 3.3.

This multi-modal feedback approach not only improves the user's control over remote operations but also enriches their sensory experience, making interactions more intuitive and effective. By transcending physical barriers and hazardous environments, teleoperation systems offer significant advantages, such as compensating for human limitations. For instance, teleoperated surgical devices can stabilise a surgeon's hand tremors, resulting in enhanced procedural outcomes. Moreover, by adjusting the scale of force between the human operator and the remote environment, teleoperation systems can deliver substantial ergonomic benefits, reducing physical strain and improving overall operational efficiency.

Nitsch and Farber's 2013 meta-analysis [17] highlights the potential of haptic interfaces to reduce excessive force, speed up task performance, and decrease handling errors in teleoperation. Though it excluded cutaneous feedback, more recent studies, such as a 2024 survey [18], demonstrate that cutaneous/tactile feedback enhances accuracy, repeatability, user experience, and reduces task time and remote environment damage.

Overview

Figure 3.1 illustrates how robotic systems have been classified based on the level of autonomy that they pursue in accordance with Sheridan's scale of levels of automation and classification [19]. The focus of this project is on direct manually controlled telerobotics with Haptic / EMS feedback.

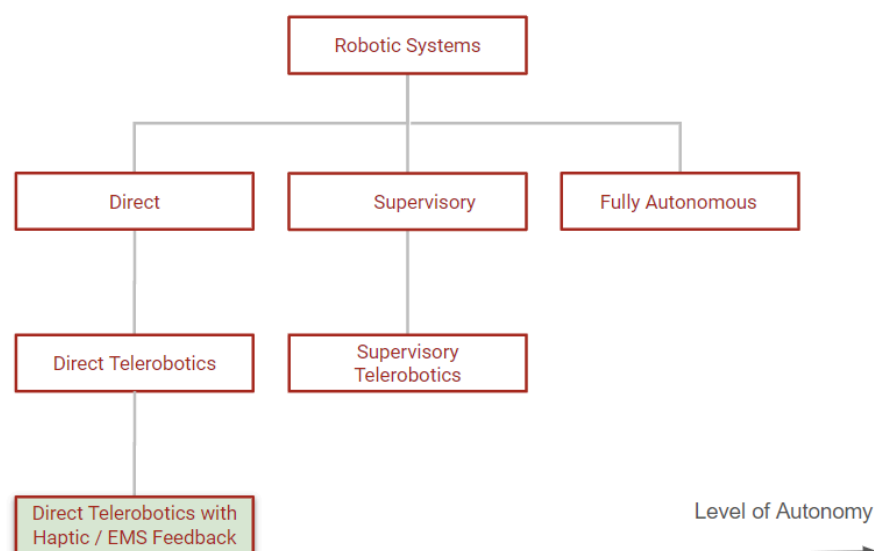


Figure 3.1: Hierarchical classification of robotic systems based on Control Mode and Levels of Autonomy with increasing level of autonomy from left to right.

After understanding the various applications of telerobotic systems, and realising the primary objectives of teleoperated systems, especially those requiring a high level of situational awareness, the rest of the section aims to cover:

- Haptic perception and feedback
- Current haptic and EMS devices
- Pseudo Haptics: visual/ auditory clues
- BCI robotics with sensory feedback

3.1 Haptic Perception and Feedback

To perceive information from our surroundings, humans are equipped with five senses: Hearing, Smelling, Tasting, Sight and Touch. Within the sensorimotor functions the haptic sense has the highest importance. It consists of a group of mechanical sensors detecting force induced deformations within tissues in the skin, muscles, and joints. Therefore, haptic perception is the sum of signals from many measurement points distributed among the human body, and these can be divided into two basic groups:

1. Tactile sensors
2. Kinaesthetic sensors

As shown in Figure 3.1.1, Tactile sensors located in exposed skin areas (eg. the fingertips) detect various stimuli such as pressure, texture, and temperature, aiding in the perception of object features like shape and edges [20]. Vibrotactile sensations, triggered by oscillatory stimuli, are sensed by specialised mechanoreceptors located subdermally [20].

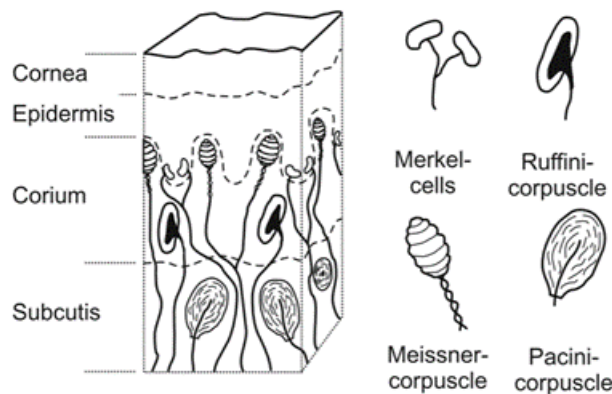


Figure 3.1.1: Distribution of sensors in the skin layer [20].

Unlike tactile sensors, kinaesthetic sensors are located mainly within muscles, joints, and tendons. They acquire forces affecting whole extremities only. These sensors help provide the sense of position and motion of one's body state [21]. Their dynamic requirements are reduced because of the mechanical low-pass characteristics of the extremities (their mass, damping and stiffness)[21].

The density of mechanoreceptors differs with the location on the body. Mechanoreceptors are denser in the glabrous skin of the hands and feet than in hairy skin, which makes touch easier to localise on the glabrous skin [22]. Additionally, haptic perception is greatly dependent on the dynamic excitation in a broad frequency range. Two distinct types of mechanoreceptors exist in humans that sense vibration, each of which is responsible for sensing a specific frequency band. Meissner corpuscles respond to lower-frequency vibrations (5–50 Hz), essential for sensing skin deformation such as when an object slips or moves under the finger [22]. In contrast, Pacinian corpuscles detect higher-frequency vibrations (40–400 Hz) resulting from transient contacts like collisions or when a tool moves across a textured surface [22].

The frequency range of haptic perception ranges from quasistatic (at least 10 s) to up to 10 kHz. The area of highest sensitivity is located within the range of 100 to 300 Hz [21]. For most practical engineering purposes, the upper range of notable sensitivity can be considered as 1kHz [21]. This is highlighted in Figure 3.1.2.

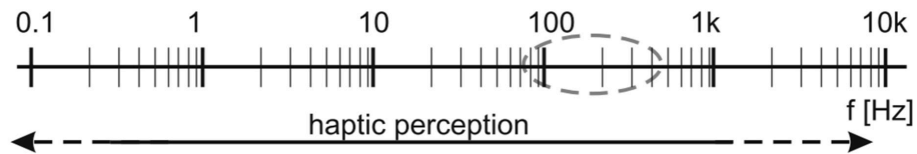


Figure 3.1.2: Range of Haptic Perception [21].

This understanding of haptic feedback perception frequency ranges and sensor types will guide the selection of appropriate feedback mechanisms for enhancing the functionality of teleoperated systems, ensuring more precise and intuitive control by operators.

3.2 Haptic Devices

Haptic devices have mechanisms for offering different forms of haptic feedback and creating various haptic illusions. These devices help create a more immersive and realistic experience for the user when interacting in the virtual world, or during teleoperation. Figure 3.2.1 showcases the classification of such haptic devices based on their wearability level.

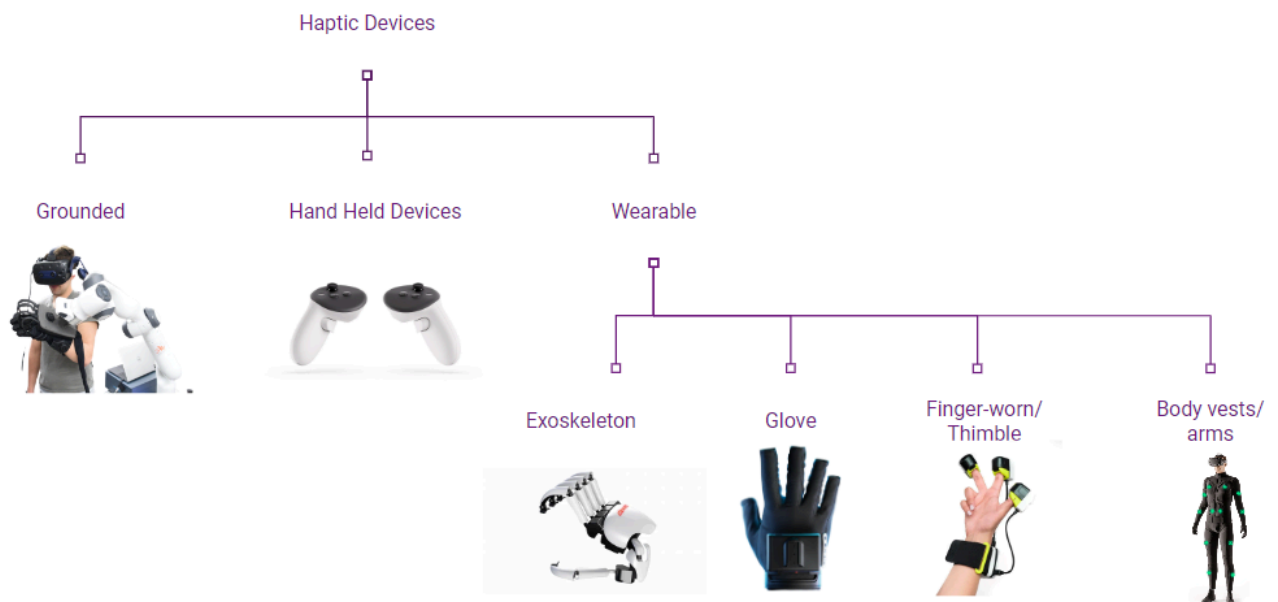


Figure 3.2.1: Haptic Device Classification based on Wearability Levels (from Left to Right: Grounded Haptic Glove [23], Quest Hand Controllers [24], Dexmo Exoskeleton [25], Manus Prime Glove [26], Weart Touch Diver Thimble [27], Tesla Haptic Bodysuit [28]).

3.2.1 Classification based on Haptic Feedback

Based on the type of feedback, haptic devices can be classified into two categories:

1. Kinesthetic Devices
2. Tactile Devices

Kinesthetic Devices

These devices target simulating the kinesthetic sensors in the skin. Stimulating these receptors can produce the illusion of movement and/or force. These devices can either be externally grounded, body grounded or ungrounded. It must be noted that ungrounded kinesthetic devices actually rely on creating tactile triggers that serve to provide the illusion of kinesthetic forces. Figure 3.2.1 demonstrates examples of a world-grounded and body-grounded kinesthetic haptic device.

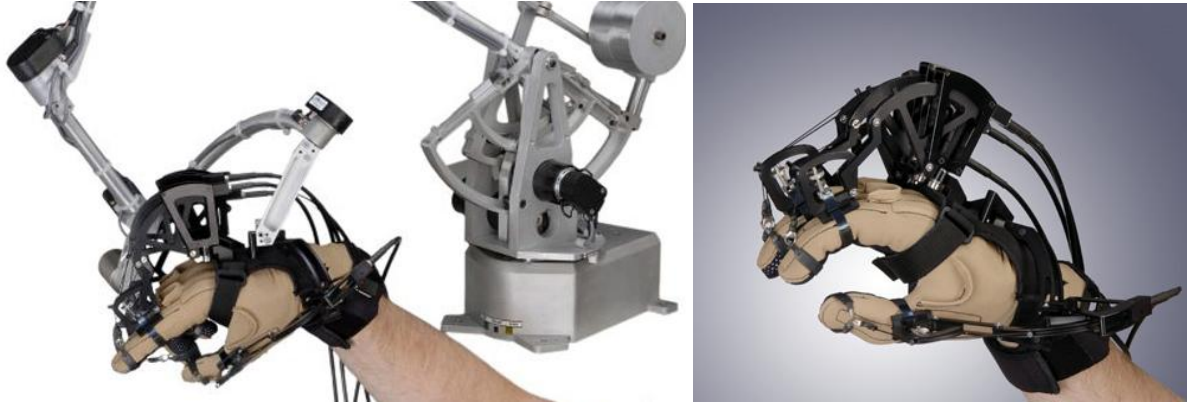


Figure 3.2.1.1: From left to right, CyberForce and CyberGrasp [29], demonstrating examples of a world (externally) grounded and a body grounded haptic device, respectively.

Tactile Devices

Tactile sensation refers to the feeling of touch, primarily perceived through the skin. The receptors involved in this process are various types of nerve endings in the skin. Unlike kinesthetic devices, which create the illusion of force and movement, tactile devices focus on simulating the sense of touch.

Typical tactile feedback provided include:

- Cutaneous Force (Skin Stretch)
- Vibration
- Thermal

Figure 3.2.1.2 depicts the Weart Touch Diver finger-worn glove. It integrates cutaneous force (skin stretch), vibrations as well as thermal feedback ranging from 15°C to 42°C with a 0.1°C precision.



Figure 3.2.1.2: The Weart Touch Diver thimble. [30]

3.2.2 Haptic Illusions and Haptic Mechanisms

Various mechanisms and designs of the haptic gloves lead to different haptic illusions like weight, stiffness and shape of the virtual object. Figure 3.2.2.1 showcases glove models with three different force feedback mechanisms: resistive force, locking mechanisms, and pneumatic actuation [31].

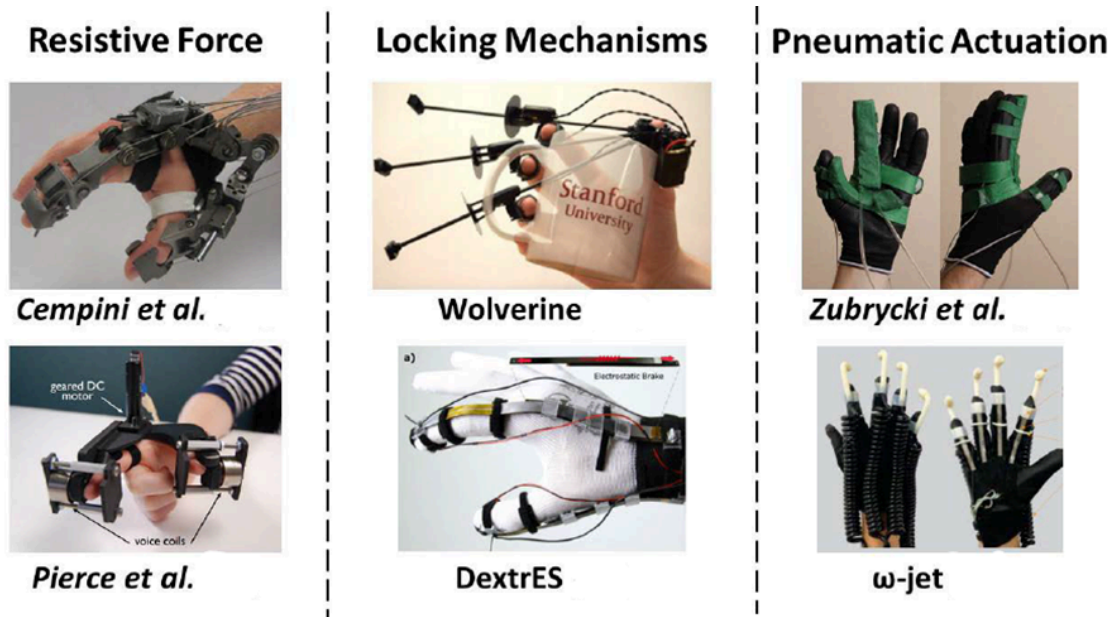


Figure 3.2.2.1: Examples of Force Feedback Glove Models based on Mechanism (from left to right: Resistive Force, Locking Mechanism, Pneumatic Action) [31].

Table 3.2.2.1 summarises the haptic illusions caused by various mechanisms as an aid for rapid prototyping or evaluation of a haptic glove based on the use case.

Table 3.2.3.1: Mechanisms Deployed for creating Haptic Illusion (one or more)[31].

Haptic Illusion	Tactile / Cutaneous Feedback				Kinesthetic Feedback		
	Vibrotactile	Skin-stretch	Compression	Electrical	Resistive Force	Locking Mechanism	Air propulsion (pneumatic)
Weight	✓	✓	✗	✗	✗	✗	✗
Shape	✗	✓	✗	✗	✓	✓	✗
Size	✗	✗	✗	✗	✓	✓	✗
Stiffness	✓	✓	✓	✗	✓	✗	✓
Texture	✓	✗	✗	✓	✗	✗	✗

3.2.3 Comparison of Common Commercially Available Haptic Glove Features

Three notable haptic gloves currently available in the market include the Weart TouchDIVER, Manus Prime 3, SenseGlove Nova, as can be seen in Figure 3.2.3.1. Information on these three gloves have been compiled in Table 3.2.3.1 so that a comparison can be made to better understand the current capabilities of haptic gloves. Some notable comments include the differences in cost, weight and the range of features available.



Figure 3.2.3.1: Weart Touch DIVER [32], Sense Glove Nova [33], Manus Prime III [34] (from left to right) .

Table 3.2.3.1: Commercially Available Haptic Gloves Comparison [32] [33] [34].

Feature	WEART	SenseGlove Nova	Manus Prime-III
<i>Force Feedback</i>			
Resistive	✗	✓	✗
Breaking	✗	✓	✗
<i>Tactile Feedback</i>			
Vibrotactile	✓	✓	✓
Skin Stretch	✓	✗	✗
Compression	✓	✗	✗
Electrical	✗	✗	✗
Thermal Feedback	✓	✗	✗
<i>Haptic Illusions</i>			
Weight	✗	✓	✓
Shape	✓	✗	✗
Size	✗	✓	✗
Stiffness	✓	✓	✓
Texture	✓	✓	✓
<i>Product Specifications</i>			
Type	Thimble	Exoskeleton	Glove
Active DoFs	6	5	9
Fingers	3	5	5
Finger Tracking	✓	✓	✓
Actuators	Electromagnetic & Thermal	Electromagnetic	Electromagnetic
Weight	200	320	150
Cost (USD)	\$3,999	\$599	\$3,999

3.3 EMS based Devices

The traditional haptic devices discussed in Section 3.2 are restricted by the area or position where it is possible for them to exert force. To counter this, controlled electrical pulses can be applied via electrode pads to induce muscle contractions [35]. This technique of electrical muscle stimulation (EMS) allows haptic feedback in a broader area or position on the human body by simply changing the electrode position. For instance, a group of researchers at A*STAR Institute for Infocomm Research developed a mixed reality tennis game that uses electrodes to simulate the weight of the tennis ball [36]. Whilst the mechanical actuators are more accurate and precise than the EMS based devices due to a number of reasons [37], EMS based devices still offer distinct advantages in certain applications. Notably, EMS devices are less bulky and more portable than their mechanical counterparts. Moreover, the feedback mechanism in EMS devices differs in the biological receptors they activate. EMS induces muscle contraction, which is sensed not only by muscle spindles but also by Golgi tendon organs, providing a unique type of sensory feedback that includes the perception of muscle stretch and tension [37]. However, users of EMS systems often experience a sense of diminished agency, which can adversely affect performance in teleoperated systems that rely solely on EMS for feedback. This reduction in perceived control can impact the effectiveness and precision required during complex teleoperation tasks, illustrating the need to compensate for it. Examples of EMS based devices include the Tesla Body Suit, which also features body tracking, making it useful for applications like gaming and potential robotic use cases.

3.4 Pseudo Haptics

Maintaining situational awareness and a sense of presence is crucial in telerobotic applications for ensuring task performance and user experience. However, traditional haptic devices, such as haptic gloves, present challenges due to their associated cost, bulkiness, and limited degrees of freedom (DoF), making them less effective in these environments. While effective at providing physical feedback, these devices often restrict user movement, require regular maintenance, and are typically confined to specific body parts, limiting their applicability. Pseudo-haptic techniques (PHTs) aim to overcome these limitations by simulating haptic sensations through primarily visual and auditory feedback, without any mechanical actuators attached to the body [38]. These techniques harness the plasticity of the human brain to create haptic illusions. Thus, PHTs have gained traction as they provide a more flexible, cost-effective, and immersive alternative for enhancing user experience and immersiveness in virtual and remote interactions.

3.4.1 Pseudo-Haptics with EMS

In a study by Ishikawa et al. [35], the integration of pseudo-haptic visual feedback with EMS resulted in a 30% increase in perceived force compared to using EMS alone. This finding is especially advantageous as one of the challenges of using EMS feedback, as discussed in Section 3.3, is maintaining the user's sense of agency. A survey by Xavier et al. (2024) [38] suggests that PHT could help mitigate the sense of agency issue associated with EMS by providing early visual cues that align with the user's actions. By enhancing the user's perception of control over their movements, PHT can diminish the perception that EMS is operating autonomously, thereby preserving the user's sense of agency. Moreover, the increased perceived force observed in the study conducted by Ishikawa et al. supports the idea that PHT can amplify haptic sensations, which, when combined with EMS, could further reduce the required EMS intensity, leading to less discomfort, further improving sense of agency, and lower power consumption.

3.4.2 Tendon Vibration and Pseudo-haptics

Recent advancements have explored the incorporation of tendon vibration to further enhance pseudo-haptic perceptions. Hirao et al. [39] demonstrated that the application of tendon vibration during free arm motion can increase the threshold for detecting discrepancies between visual and physical motion by approximately 13%, thereby allowing for a wider range of visual motion gains without users noticing any mismatch between the visual and physical experiences. This finding suggests that tendon vibration not only serves as an additional haptic cue but also acts as a filter, effectively masking minor inconsistencies in haptic feedback and thereby enhancing the overall effectiveness of pseudo-haptics. The study additionally shows that tendon vibration can amplify the perception of weight, providing a haptic cue that complements visual feedback and results in more nuanced and realistic haptic sensations.

PHTs can be integrated with traditional haptic devices like haptic gloves to improve user experience and boost task performance. With this, synchronisation of haptic and visual cues could become an issue. The study by Hirao et al. [39] offers a potential solution by demonstrating that tendon vibration can help alleviate these synchronisation issues to a certain extent, ensuring a more cohesive and immersive haptic experience for the user. On the other hand, while multimodal feedback, including pseudo-haptics, can enhance user immersion and situational awareness, it may also lead to elevated cognitive load. Studies by NASA [40] indicate that elevated cognitive load and loss of situational awareness can adversely affect task performance during teleoperation. Notably, the study by Xavier et al. [38] found that visual PHT was more effective than both sound and vibration feedback in controlling grasping force, while maintaining a lower cognitive load, thereby making visual PHT a valuable cognitive aid in tasks requiring situational awareness.

3.5 Mind Controlled Robotics

Robotic arms controlled by Brain-Computer Interfaces (BCIs) represent a significant advancement in assistive technology, especially for individuals with severe mobility restrictions such as those with tetraplegia, advanced neuromuscular disorders, or amputations [41]. BCIs establish a direct communication pathway between the user's brain and external devices, bypassing traditional neuromuscular routes. This is done with the use of Electroencephalography (EEG). It measures and records electrical signals through sensors arrayed across the scalp. These signals are then processed and translated into commands that control external devices like prosthetics, enabling applications in gaming, virtual reality, and robotics [42]. An example of this innovative application is the Hand Exoskeleton for Rehabilitation Objectives (HERO) project [43], which integrates EEG-controlled commands to operate a hand exoskeleton. This exoskeleton aids in the rehabilitation of stroke survivors by allowing them to perform motor tasks with their hands using thoughts alone. The 2007 study by Chatterjee et al. [44] provides an insightful demonstration of using vibrotactile feedback as an effective biofeedback modality for BCIs, improving user control in neuroprosthetic applications through enhanced sensory interaction.

Building on these advancements, existing research has also demonstrated the feasibility of using a non-invasive EEG cap to select target goal positions for the movement of a teleoperated robotics system [45]. Furthermore, consumer products such as Galiea (Figure 3.5.1) and Nextmind Dev Kit show the potential in using a gamified calibration process to train the system in recognising the patterns in EEG signals (Figure 3.5.2).



Figure 3.5.1: Galea EEG VR Integrated Headset [46].

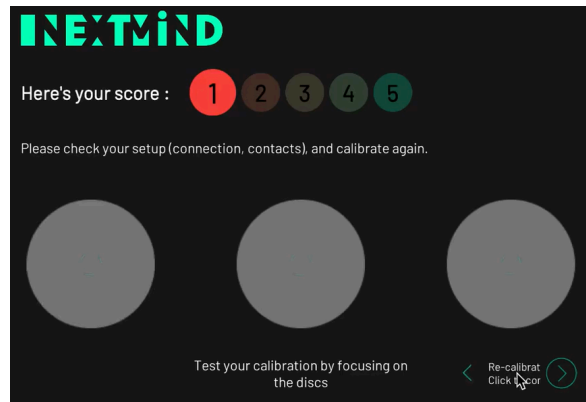


Figure 3.5.2: NextMind Dev Kit's Gamified Calibration Process [47].

3.6 Closing Remarks

This section has attempted to establish a foundational understanding and knowledge for working with telerobotic systems. It identified four key objectives in telerobotic systems, namely maintaining robustness, sense of presence, task performance, and operational efficiency [16]. It discussed how integrating haptic feedback and Head-Mounted Displays (HMDs) addresses these objectives, thereby enhancing the effectiveness and user experience of telerobotic operations. The exploration of haptic perception and feedback in Section 3.1 laid the groundwork for understanding the various haptic devices and technologies available. Section 3.2 presented a classification of haptic devices based on the wearability level as well as the type of haptic feedback. It included a concise guide to aid in the prototyping and haptic-illusion capabilities of haptic gloves. The comparison of industrially available haptic gloves (Weart TouchDIVER, Sense Glove Nova, Manus Prime III) provided insights into current market offerings and their capabilities. In the next section, the discussion on Electrical Muscle Stimulation (EMS) feedback highlighted its potential to overcome limitations of traditional haptic devices, offering a broader range of feedback possibilities. However, it was noted that EMS systems can lead to a diminished sense of agency, which needs to be addressed for effective teleoperation. In Section 3.4, pseudo-haptic techniques (PHTs) were presented as a promising approach to enhance user experience and immersion while potentially reducing costs and overcoming limitations of traditional haptic devices. It was noted that they have the potential to retain the user's sense of agency while using EMS feedback. It is important to consider that while multimodal feedback techniques add to the user's immersiveness and make the experience more realistic, it can also lead to high cognitive load. Beyond a certain threshold, this overload may negatively impact task performance [40], hence, care must be taken while integrating additional modes of feedback. Moreover, it remains challenging to know what combination of haptic feedback is the best mode of conveying feedback to the user for enhanced performance [18]. Additionally, it was discovered that adding tendon vibrations to PHTs increases the threshold for detecting discrepancies between visual and physical motion [39]. Finally, a brief exploration was presented in the field of BCI arm robotics with sensory feedback, for the stretch-goal portion of this project.

4 Requirements, Specifications and Approach

4.1 Requirements

Table 4.1.1 to Table 4.1.5 detail the project requirements in five distinct categories.

Table 4.1.1: High level requirements.

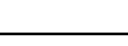
Requirement ID	Requirement Description	Client Approval
[H.001]	Users shall be able to control a robotic arm in virtual reality with hand and arm haptic feedback.	
[H.002]	The VR headset should have a minimum refresh rate of 90 Hz to minimise the risk of cybersickness.	
[H.003]	The system must allow for accurate tracking of finger and hand movements through the glove and VR headset to allow for synchronisation with the robotic arm's actions.	
[H.004]	The VR interface shall be intuitive and immersive for controlling the arm, adjusting settings and monitoring feedback.	
[H.005]	The system must have a simple calibration procedure (maximum 2 minutes) to align the glove, haptic suit and VR headset with the robotic arm for accurate tracking and feedback.	

Table 4.1.2: Financial requirements.

Requirement ID	Requirement Description	Client Approval
[F.001]	The construction of the prototype haptic feedback glove must not add more than \$450 in additional cost.	

Table 4.1.3: Safety requirements.

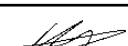
Requirement ID	Requirement Description	Client Approval
[S.001]	The system should have features to prevent physical harm or strain of the operator.	
[S.002]	Apply appropriate safety measures to ensure the robotic arm does not collide with itself or the environment.	

Table 4.1.4: Physical requirements.

Requirement ID	Requirement Description	Client Approval
[P.001]	A minimum 2-finger haptic feedback glove must provide kinesthetic and vibrotactile feedback to the user.	
[P.002]	The haptic feedback glove should not weigh more than 500 g.	
[P.003]	The haptic feedback glove shall be adjustable to accommodate various hand sizes and shapes.	
[P.004]	The system should be operable within the temperature range of 10 to 30 degrees celsius which exceeds the maximum expected temperature fluctuation in the controlled robotics lab environment.	
[P.005]	The system must be untethered using in-built batteries or a portable charger, capable of providing a minimum of 10W (5V at 2A).	
[P.006]	The wearable components of the system should be sweat resistant to perform continuously in situations where the user may be sweating heavily from physical activity. The device should be able to handle solid particles and water according to IP41 or better.	

Table 4.1.5: Non-Functional Requirements.

Requirement ID	Requirement Description	Client Approval
[N.001]	The haptic glove created should use readably available commercial off-the-shelf components and be designed to optimise maintainability and replacement of hardware.	
[N.002]	Supporting documentation of how to use and maintain the system should be created, for an audience of technically competent engineers.	
[N.003]	The devices should be compatible with OPC Unified Architecture to integrate with the data lake at the MSM lab.	
[N.004]	The software must be developed in Unity with relevant integration to the hardware which may use arduino, python and C++.	
[N.005]	The software and hardware should be easily expandable, allowing for deployment with more units, additional hardware integration or foreseeable expanded functionality.	
[N.006]	An existing robotic arm, provided by the supervisor, must be used for the project.	

4.2 Design Specification & Approach

4.2.1 Proposed Project Architecture

Relevant Requirements: [H.001] [N.002] [N.003] [N.004] [N.005] [S.002]

The preliminary diagram, in Figure 4.2.1.1, outlines the connections between the components required to achieve VR teleoperation with haptic feedback [H.001], it will be updated as the project progresses and supplement more detailed user manuals within the project documentation [N.002]. Key integrations required for the project are communication with the existing MSM DataLake over OPC UA [N.003] along with hardware integration through pre-existing packages with unity [N.004].

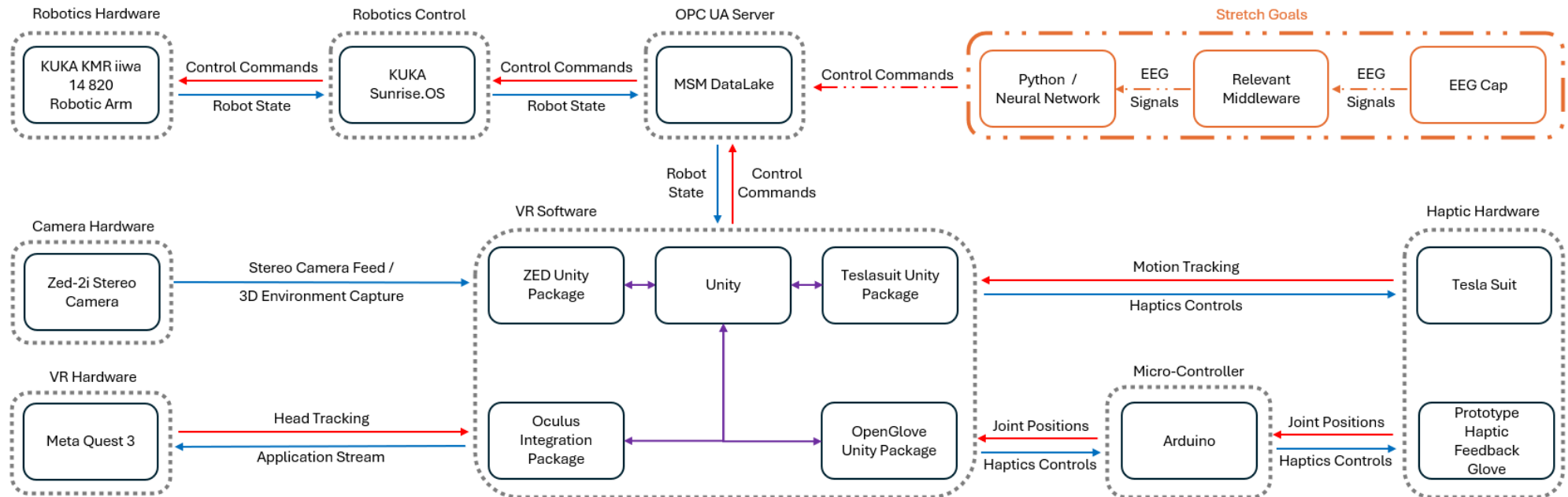


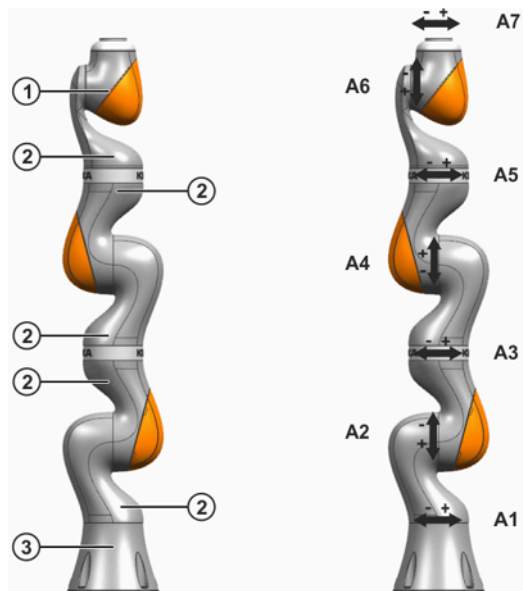
Figure 4.2.1.1: Proposed Project Architecture.

All code will be sufficiently documented and designed to be scalable irrespective of the number of active VR users in the environment [N.005]. Control commands sent to KUKA's Sunrise.OS will only execute if the target is within the specified workspace preventing the robot from breaching safety zones or encountering singularities [S.002].

4.2.2 KUKA Robotic Arm in MSM Lab

Relevant Requirements: [H.001] [N.006] [S.002]

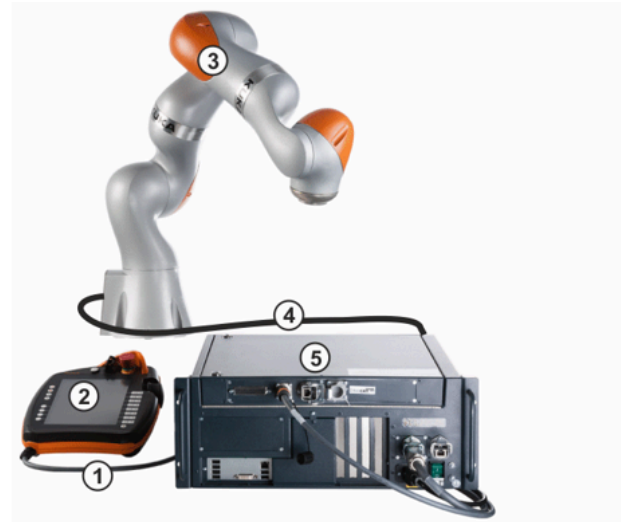
The robotic arm to be used for teleoperation is the 7-DOF iiwa 14 R820 (Figure 4.2.2.1) present in the MSM Lab [N.006]. Furthermore, a trained supervisor must be physically present and in control of the KUKA smartPAD during VR teleoperation. The smartPAD's E-stop, shown in Figure 4.2.2.2, will provide an additional layer of safety in ensuring the robot cannot cause harm to itself or the surrounding environment [S.002].



Main assemblies and robot axes

- 1 In-line wrist
- 2 Joint module
- 3 Base frame

Figure 4.2.2.1: Robotic Arm Join Breakdown [48].



Overview of robot system

- 1 Connecting cable to the smartPAD
- 2 KUKA smartPAD control panel
- 3 Manipulator
- 4 Connecting cable to KUKA Sunrise Cabinet robot controller
- 5 KUKA Sunrise Cabinet robot controller

Figure 4.2.2.2: KUKA smartPAD Configuration [48].

The arm is mounted to a mobile KMR unit as seen in Figure 4.2.2.3; however, the unit will be stationary as moving the platform is outside the project scope. Existing integration with the MSM Lab server will provide updates regarding the robot's state for haptic feedback [H.001].



Figure 4.2.2.3 KUKA Robotic Arm with the Mobile Unit in MSM Lab.

4.2.3 ZED 2i Stereo Camera

Relevant Requirements: [H.001] [H.004]

Utilising depth and position data, the ZED 2i shown in Figure 4.2.3.1 offers spatial mapping (3D reconstruction) through the ZED SDK [49]. StereoLabs provides the required software functionality to generate a single real-time texture map which can be used to update the user's view of the 3D environment in Unity [H.001].



Figure 4.2.3.1: ZED 2i Stereo Camera [49].

However, if the results with this approach are deemed unsatisfactory, a simpler approach of streaming the stereo camera feeds through Unity will also create an immersive environment [H.004].

4.2.4 Open Source Haptic Feedback Glove Design LucidVR 4.1

Relevant Requirements: [H.003] [H.005] [F.001] [P.001] [P.002] [P.003] [P.006] [N.001] [S.001]

The existing LucidVR V4.1 glove design, seen in Figure 4.2.4.1, offers kinesthetic feedback and motion tracking (using potentiometers) [H.003] for five fingers with built in calibration functionality [H.005] and integration with SteamVR through OpenGloves. The MG90s servos provide a maximum of 1.8kg/cm (0.177Nm) at 4.8V [50], far below the level required to harm human fingers [S.001].

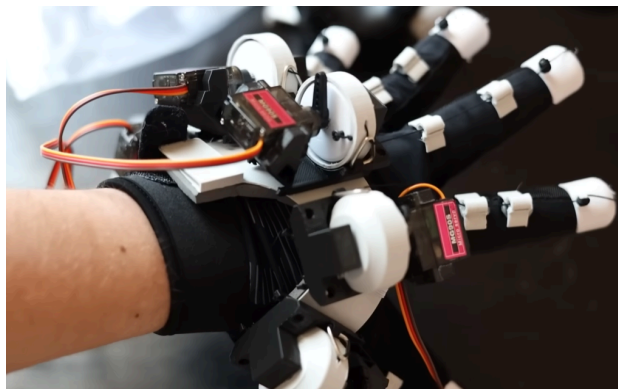


Figure 4.2.4.1: Open Source LucidVR V4.1 Haptic Feedback Gloves [51].

The LucidVR glove design will be used as the basis for a modified two finger variant aiming to also provide vibrotactile feedback [P.001]. The costs of the modified projected part list detailed in Table A.1 in Appendix A: Proposed Prototype Haptic Glove Parts List do not exceed \$100 per hand for a combined total less than the maximum additional project expenditure of \$450 [F.001].

The SenseGlove-Nova (a comparable commercial glove with vibrotactile and kinesthetic feedback) weighs 320 grams [52], an additional 180g allowance has been provided (for a total of 500g per glove) [P.002] given the project limitations of using off the shelf hardware which may not be as weight efficient [N.001].

The 3D printed housings will be designed to ensure a rating of IP41 or higher is achieved [P.006]. The first digit 4, implies the glove components must be protected against the ingress of foreign particles > 1mm. The second digit, 1 requires that vertically falling water droplets shall have no harmful effect on the system's operation [53].

While a medium sized base glove will be used as the base, the system is modular and can easily be recreated for different sized gloves [P.003].

4.2.5 Haptic Feedback Glove Power Supply

Relevant Requirements: [P.005]

A commercially available power bank, such as the example in Figure 4.2.5.1, can provide 22.5W, which is sufficient to power the haptic feedback gloves and allow the system to operate untethered [P.005].



Figure 4.2.5.1: Example commercially available 10,000mAh power bank [54].

4.2.6 Tesla Haptic Feedback and Motion Tracking Suit

Relevant Requirements: [H.001] [H.003] [H.005] [P.004] [P.005] [P.006] [S.001]

While only the Tesla suit's haptic feedback jacket is relevant to the scope of this project, it contains 48 channels which control 62 distinct EMS/TENS electrodes (Figure 4.2.6.1) to provide haptic feedback to the upper body including the arms [H.001]. Accurate tracking of hand movement may be achieved using the suit's 14 individual 9-axis IMUs (Figure 4.2.6.2) [H.003] which will also serve to calibrate the system on startup by requesting users assume a T-pose [H.005].

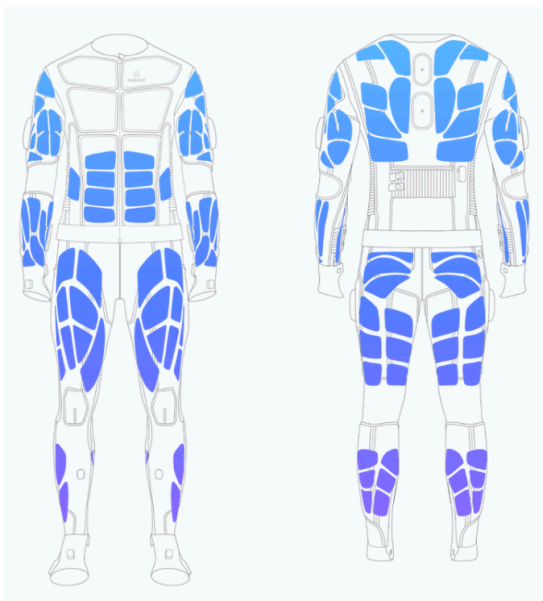


Figure 4.2.6.1: Tesla Suit Distinct EMS/TENS Regions [55]

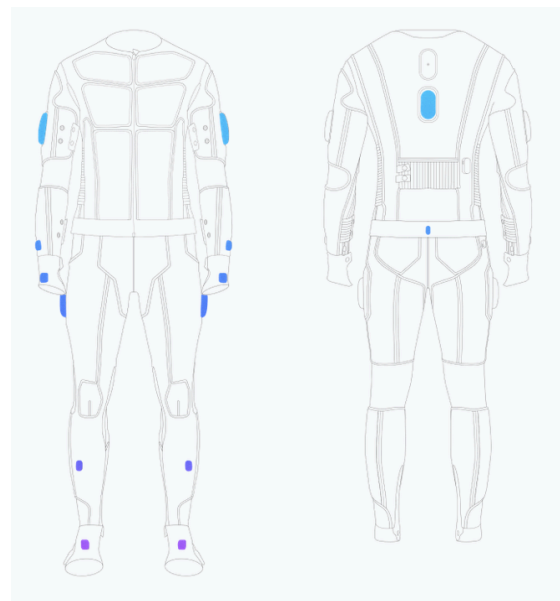


Figure 4.2.6.2: Tesla Suit Motion Tracking Modules [55]

The suit has a safe operating range of 5°C to 35°C which exceeds the system requirements [P.004]. An inbuilt 10,000 mAh battery and wireless connectivity will allow the suit to operate untethered [P.005]. While no official IP rating is provided by Teslasuit, the bottom layer in contact with the user is sweat resistant and composed of 55% nylon with 45% spandex [P.006].

Furthermore, an in-built heart rate monitoring system will allow the system to detect exertion and provide alerts in the VR interface to prevent excessive user strain [S.001].

4.2.7 Meta Quest 3

Relevant Requirements: [H.002] [H.004] [H.005] [P.005] [N.005] [S.001]

Table 4.2.7.1 displays a summary of the key factors which contributed to the choice of VR headset.

Table 4.2.7.1: Comparison of screen resolution and cost for available VR headset options.

Headset	Meta Quest 2 [56] 	Meta Quest 3 [57] 	Apple Vision Pro [58] 
Per-eye Screen Resolution	1832×1920p	2064×2208p	3,660 × 3,200p
Unit Cost	\$440	\$799.99	\$5,999

To maximise immersion [H.004], the Meta Quest 3 VR headset was chosen over the Quest 2 due to its roughly 30% higher per-eye screen resolution [59]. While the Apple Vision Pro offered an even higher per-eye screen resolution, the Quest headsets were preferable due to integration limitations with Apple’s closed ecosystem. Furthermore, the limited availability of Vision Pro headsets in Australia and their higher relative cost would have negatively impacted the scalability of the system [N.005].

The Quest 3’s default refresh rate of 90Hz meets the requirement to minimise the risk of cybersickness [H.002] with the potential to be increased to 120 Hz. In the setup process, the Quest’s boot-up calibration will match the environment to the user’s height which will work in conjunction with the other system calibration steps [H.005]. With the in-built battery and “Air Link” connection protocol the headset will be able to operate untethered [P.005]. Furthermore, the Quest’s built-in “boundary” functionality will detect obstacles and display the user’s real world environment in pass-through mode supplementing our safety precautions by protecting the user from collisions with the environment [S.001].

4.2.8 Computational Needs

Given the high computational load required for VR development, the project is to be conducted on high end consumer PCs equipped with NVIDIA RTX-4060-Ti graphics card. As such, the plan is to utilise the resources available at the Digital Twin Labs and those provided by Monash Nexus for Emerging Technologies (MNET).

5 Scope, Project Plan & Timeline

5.1 Project Scope

The objects and tasks that are in-scope and out-of-scope are clarified in Section 5.1.1 and Section 5.1.2 of the report, respectively.

5.1.1 In Scope

- Developing and implementing VR headset control for real-time manipulation of the robotic arm.
- Designing and building a haptic glove to provide users with real-time feedback through touch.
- Integrating the jacket of a haptic suit to enhance user awareness of the robotic arm's motion constraints.
- Ensuring seamless interaction between the haptic glove, suit, VR headset, and robotic arm.
- Assessing how the immersive features impact operator performance and cognitive strain.
- Conducting research and development into brain signal-based control mechanisms, including hardware acquisition and neural network training.

5.1.2 Out-of-Scope

- Controlling the movement of the mobile platform to which the robotic arm is mounted (KMR iiwa mobile platform).
- Applying haptic feedback to the user's legs and lower body.
- Developing or inventing new types of haptic feedback technology beyond existing research and commercial solutions.
- Developing algorithms that adjust haptic feedback based on user behavior or specific task requirements.
- Implementing new functionalities for the robotic arm unrelated to enhancing VR control or haptic feedback.
- Integrating more than two distinct haptic feedback mechanisms into the glove and more than two pseudo-haptic feedback features into the VR system.
- Conducting large-scale or long-term studies on the impact beyond the initially planned scope.
- Using advanced techniques such as Gaussian Splatting or NeRF for scene reconstruction from multiple 2D camera views.
- Developing features for multi-user interaction within the VR environment.
- Investigating how different object material properties (e.g., soft, rigid, fragile) can be more accurately represented in haptic feedback.
- Investigating how various user demographics interact with haptic feedback systems and adapting designs accordingly.
- Implementing feedback mechanisms for varying textures or nuanced sensations, like weight.

5.2 Project Plan & Timeline

Figure 5.2.1 illustrates a high-level summary of the project timeline. For a more detailed Gantt Chart which further breaks down each of the sub-categories, please refer to Appendix B: Project Timeline. The project is divided into 9 major sub-categories as can be seen in Figure 5.2.1. The 9 major sub-categories include:

1. Project Proposal
2. Progress Report
3. Integration of Robotic Arm Control via VR Headset
4. Development of a Haptic Glove
5. Implementation of a Haptic Suit
6. Integration of Haptic Glove and Suit with VR headset and Robotic Arm

7. Evaluating the Benefits of Immersive Feedback on Operators
8. Stretch-Goal - Mind-Controlled Robotic Arm Operation
9. Final Report Submission

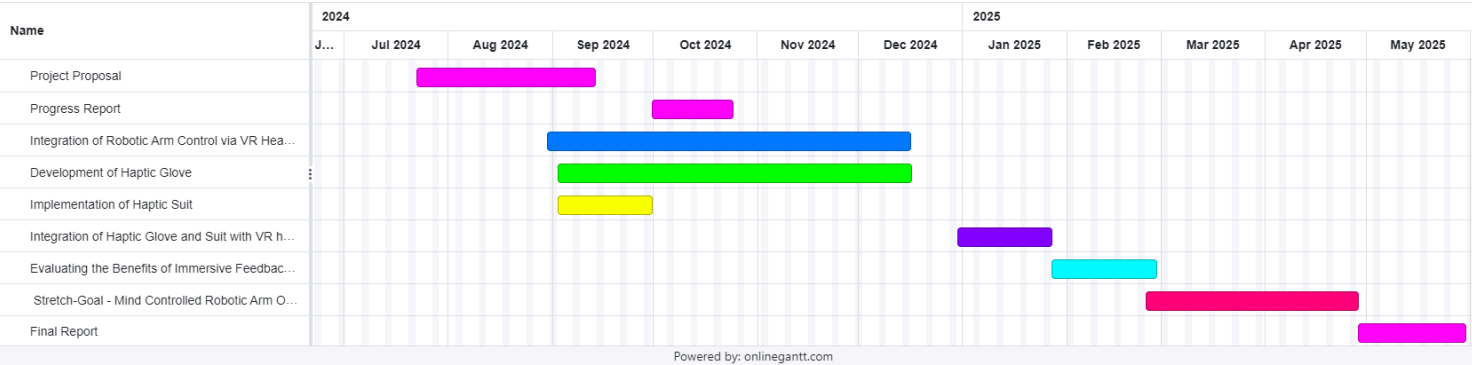


Figure 5.2.1: High-Level summary of project timeline. For a more detailed breakdown, please refer to Appendix B: Project Timeline.

As illustrated in Figure 5.2.1 and Appendix B: Project Timeline, after the project proposal submission, there will be approximately four weeks before the drafting of the progress report begins. During this period, the team plan to initiate work on the following three main tasks:

- Integration of Robotic Arm Control via VR Headset
- Development of a Haptic Glove
- Implementation of a Haptic Suit

The final task involves procuring all the necessary components for the pseudo-haptic hardware and haptic glove, ensuring sufficient lead time as students complete their exams. The team has committed to working through the summer holidays to integrate the three systems and assess the benefits of the immersive features.

Upon returning from the break, the goal will be to develop and implement the mind-control robotic arm operation as a stretch goal for Project B. This will be followed by adequate time to produce a high-quality final report, poster, and video. If unforeseen issues arise during the summer break, the stretch goal may be adjusted or cancelled, allowing more time for integration and testing.

The detailed Gantt Chart seen in Appendix B: Project Timeline accounts for sufficient time for product purchasing, integration and testing, finalising reports, videos, and posters, and obtaining feedback from relevant supervisors.

5.3 Work Breakdown

To clearly define each individual's role and corresponding responsibilities, Table 5.3.1 and Table 5.3.2 (respectively) provide detailed information. To guarantee that work is evenly distributed across the project, Figure 5.3.1 presents a summary of the agreed-upon task allocation among team members. Each of the three team members has a general team responsibility, as outlined in Appendix C: Team Contract, and is also assigned specific involvement for different parts of the project, detailed in Figure 5.3.1. For further information on the team's communication, management, and decision-making processes, please refer to Appendix D: Team Communication and Management Agreement.

Table 5.3.1: Table outlining the responsibilities of each person involved in this project.

Full Name	Contact Details	Roles
Jason, Abi Chebli	jabi0003@student.monash.edu	Team Member
Taha, Samani	tsam0009@student.monash.edu	Team Member
Aarushi, Raheja	arah0033@student.monash.edu	Team Member
Keenan, Granland	Keenan.Grandland@monash.edu	Project Stakeholder
Chao, Chen	Chao.Chen@monash.edu	Supervisor
Rohan, Kalanje	rohan.kalanjerajesh@monash.edu	MNET Supervisor

Table 5.3.2: Table outlining the different roles and responsibilities.

Role	Responsibility
Team Member	● Actively becomes 'Chairperson' or 'Minute Taker' for each meeting ● Work on tasks assigned and agreed upon as seen in Figure 5.3.1 ● Prepare for meetings according to agenda ● Participate in discussions actively during meetings
Project Stakeholder	● Supervises and manages teamwork done in the Smart Manufacturing Lab ● Check the work done by members to ensure the team is on track ● Resolves any conflicts that may arise (as agreed upon in Appendix C: Team Contract) ● Helps connect the team to other talent to support their project
Supervisor	● Supports the team on a high-level ● Ensures that the general direction of the teamwork is appropriate ● Helps connect the team to other talent to support their project
MNET Supervisor	● Helps provide access to rooms and Full Body Haptic Teslasuit ● Helps connect the team to other talent to support their project

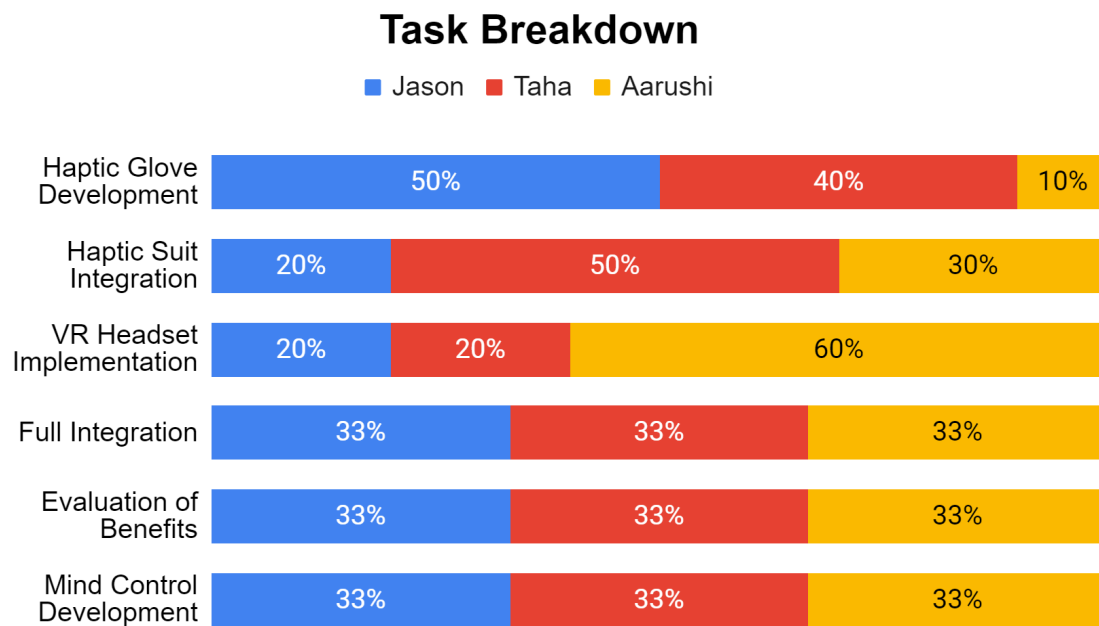


Figure 5.3.1: Figure outlining the distribution of each member's involvement in the different tasks and stages of the project.

6 Risk Management Plan

The Occupational Health and Safety (OHS) risks were considered and registered using Monash University's Risk Management system, Safety and Risk Analysis Hub (SARAH). A condensed printout of the Risk Assessment completed in SARAH can be found in Appendix E: Project Risk Assessment. For a less condensed version of the Risk Assessment, please visit the Monash University SARAH platform.

For non-OHS project risks, the Risk Assessment Matrix used can be seen in Table 6.1, while a Risk Assessment Table for non-OHS project risks can be seen in Table 6.2.

Table 6.1: Risk Assessment Matrix used.

Likelihood	Consequences				
	Insignificant <i>Risk is easily mitigated by normal day-to-day processes</i>	Minor <i>Up to 10% delay in overall project timeline.</i>	Moderate <i>10% to 30% delay in overall project timeline.</i>	Major <i>30% to 50% delay in overall project timeline.</i>	Catastrophic <i>Over 50% delay in overall project timeline.</i>
Almost Certain >90% Chance	Medium	High	High	Extreme	Extreme
Likely 50%-90% Chance	Medium	Medium	High	High	Extreme
Possible 10%-50% Chance	Low	Medium	Medium	High	High
Unlikely 3%-10% Chance	Low	Low	Medium	Medium	High
Rare <3% Chance	Negligible	Low	Low	Medium	Medium

Table 6.2: Risk Assessment Table for foreseeable non-OHS Project Risks.

Project Risk	Risk	Likelihood	Consequence	Risk Level	Mitigation	Residual Risk
Delayed delivery of project components for Haptic Glove and Pseudo Haptic Features	Delayed construction and implementation of haptic glove and pseudo-haptic features.	Possible	Moderate	Medium	Make a list of all components needed for the project and order them ASAP. Discuss with the supervisor to identify alternative components or sources.	Components are still not available. Utilise resources available on campus to design around and construct the prototype instead.
3D printer filament and e-waste generated and disposal	Environmental impact due to mishandling of 3D printer filament waste and e-waste.	Possible	Insignificant	Low	Induct all team members to understand the different methods of disposal and different types of bins available, utilising resources such	3D printed filament waste and e-waste are still not disposed of correctly, resulting in environmental impact.

					as Monash Precious Plastics.	
Lack required technical skills in this domain.	The workload will pile up and deadlines will not be met.	Possible	Moderate	Medium	Team members should proactively research online, seek assistance from manager, supervisor and others in the field. Discuss with the supervisor the best strategy to overcome this hurdle.	The work may be completed, but at a slight delay.
Improper allocation of the work among the team members	Few members may get an excessive amount of work, resulting in potentially overworked, lower quality work and lack of cohesion among the team.	Possible	Minor	Medium	Consider the strengths and weaknesses of each team member, and allocate work and assign workload difficulty using the job list. Team members should proactively inform the team when they feel an unfair workload distribution and should discuss this with the team and/or manager.	Few team members are still left completing more parts of the work. This should be reflected in some sort of individual peer review feedback system.
Miscommunication between the team members, managers and/or supervisors.	Deviation from what the manager and supervisor expect the final outcome to be, causing dissatisfaction and potential request to re-do work.	Possible	Moderate	Medium	Communication between team members should be clearly outlined in meeting minutes, with regular meetings and progress updates to track progress on tasks to avoid any major misunderstanding.	Less likely for major miscommunication to occur. For minor miscommunication or questions, clarify any uncertainty with team members, managers and/or supervisors off-line through agreed upon written communication median.
Major requirements changes after the project proposal	During the development of the project, any requirements changes might result in not getting all the agreed upon work done by the deadlines set.	Rare	Major	Medium	Frequently meet with the manager and supervisor to ensure project alignment. Get the major to sign to the agreed upon requirements. Provide updates and keep the objectives clearly defined and organised.	Less likely for any major requirement changes to occur. For minor requirement changes, clearly define the functional, non-functional requirements, what would be in-scope and out-of-scope, timeline and cost and evaluate the feasibility of making this change with the manager and supervisor.

7 References

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Appendices

Appendix A: Proposed Prototype Haptic Glove Parts List

Table A.1 is a summary of the costs associated with building the proposed two finger variant of the open source LucidVR gloves.

Table A.1: Cost Breakdown of Proposed Prototype Haptic Glove with Vibrotactile and Kinesthetic Feedback

Component	Pack Size	Pack Cost	Unit Cost	Source	Required Quantity	Net Cost (AUD)
Potentiometer 10k Ohm	10	\$24.95	\$2.50	Link	4	\$9.98
MG90S Servo Motor	10	\$31.35	\$3.14	Link	4	\$12.54
ESP-WROOM-32	1	\$11.99	\$11.99	Link	2	\$23.98
Retractable Badge Reel	10	\$8.84	\$0.88	Link	4	\$3.54
Cycling Glove Pair Medium Full Finger	1	\$11.79	\$11.79	Link	1	\$11.79
Flat Vibration Motor	10	\$10.69	\$1.07	Link	4	\$4.28
10,000mAh Power Bank	1	\$20.98	\$20.98	Link	2	\$41.96
JST-XH Connector Kit 2.54mm	1	\$22.95	\$22.95	Link	1	\$22.95
PLA+ Roll for 3D Printed Parts	1	\$23.59	\$23.59	Link	1	\$23.59
Total Cost for Both Gloves						\$154.60
Total Cost per Glove						\$77.30

To access the live Project Gantt Chart, please follow this [link](#).

Monash University, Final Year Project

Gantt Chart Template © 2006-2018 by Vertex42.com.



Monash University, Final Year Project

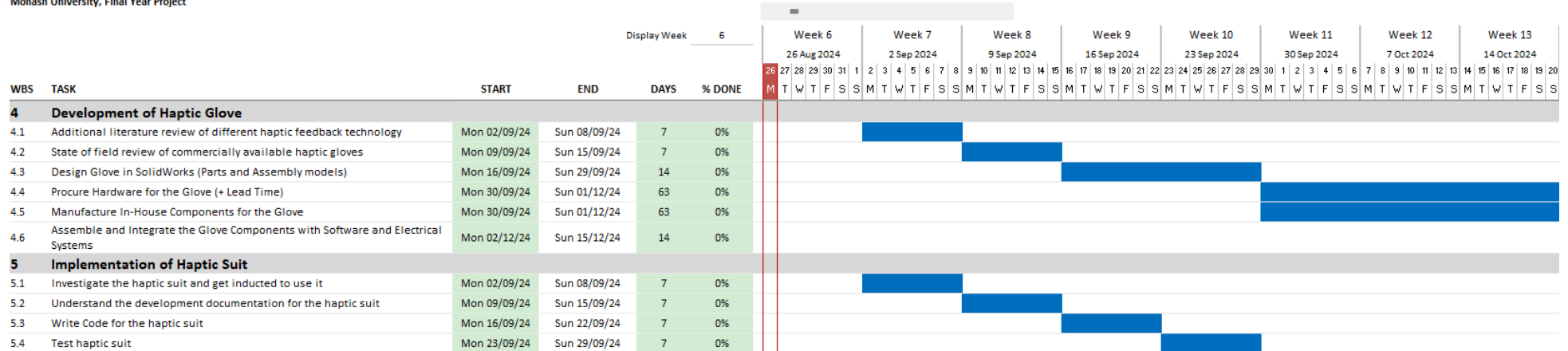
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Immersive Telerobotics with Haptic Feedback

Monash University, Final Year Project

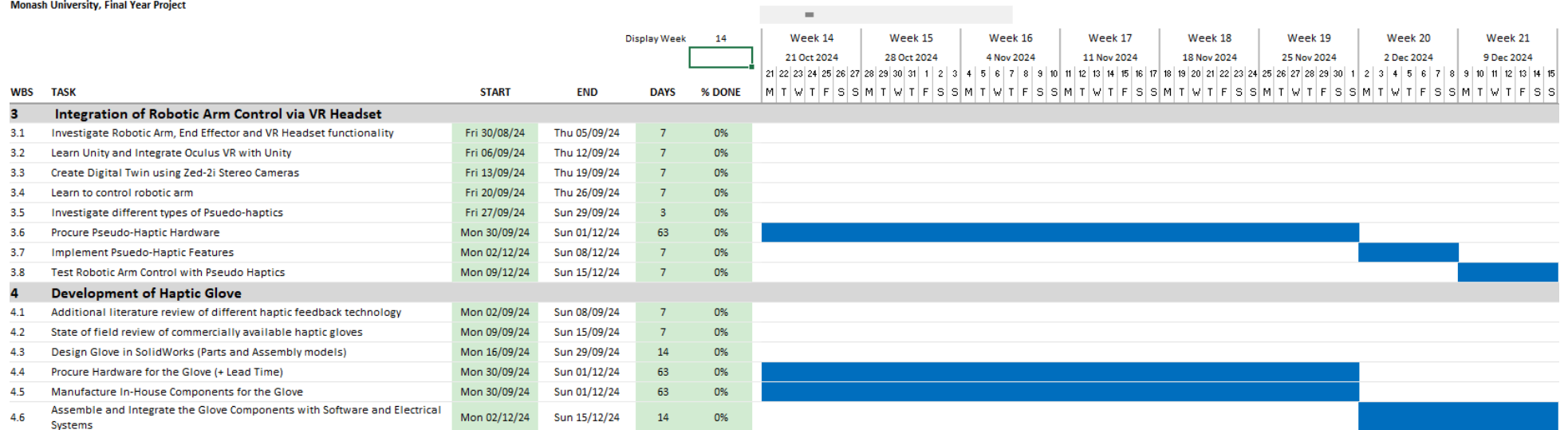
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Immersive Telerobotics with Haptic Feedback

Monash University, Final Year Project

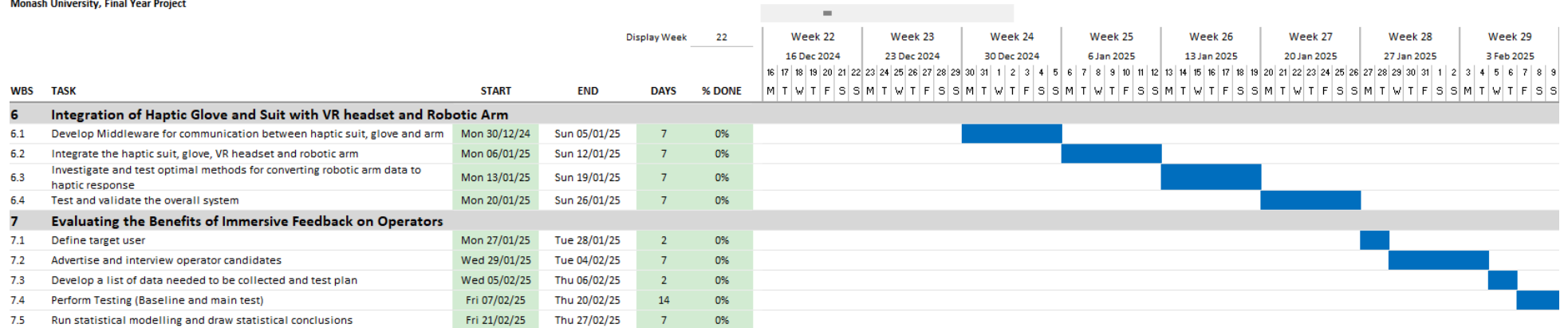
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Immersive Telerobotics with Haptic Feedback

Monash University, Final Year Project

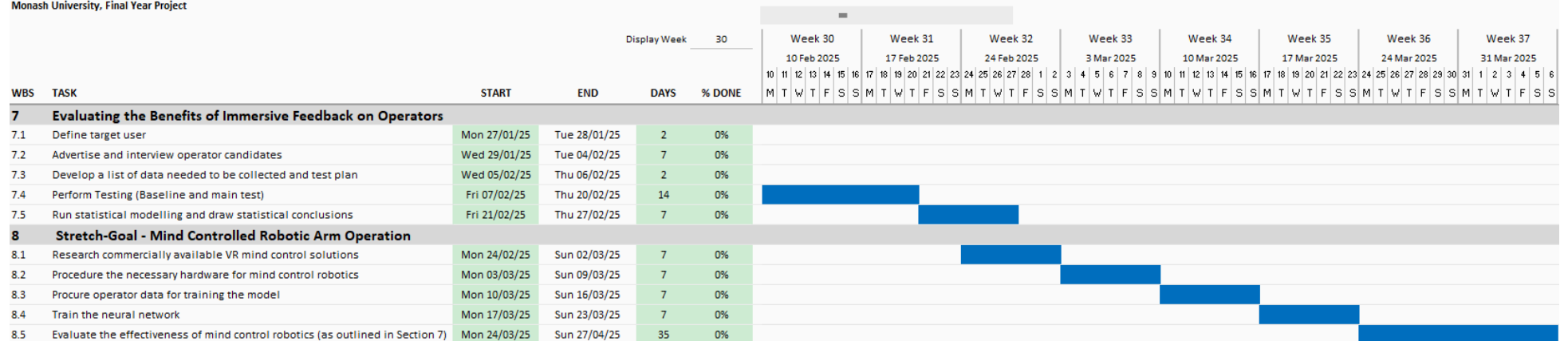
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Immersive Telerobotics with Haptic Feedback

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11

Display Week

Appendix C: Team Contract



Team Contract

Team Name: NoHaptX

Team Member Names:

Jason Abi Chebli

Taha Samani

Aarushi Raheja

1. Document Purpose

The purpose of this team contract is to outline the standard operating practices and team norms of the above named team and individually listed members for the remaining duration of the team lifespan. The guidelines outlined in this document are agreed to by all team members as indicated by their signature at the end of the contract. Any amendments to the contract must be discussed and agreed to by all signing members. Failure to abide by the outlined standard operating practices of this contract could harm the team's overall functioning and result in penalizing action as detailed in the contract.

2. Rules and Regulations

The team agrees to the following guidelines regarding general procedures, practices, and behaviours that are deemed acceptable.

A. Expectations

i. *Project Expectations*

- All team members are committed to achieving a HD in these capstone units (FYP-A and FYP-B).
- We agree to finalise an initial draft at least one week in advance for whole team review with supervisors and have a finalized version ready to submit at least two days before submission to minimise stress.
- Self-imposed deadlines have a buffer of +2 days, otherwise, special consideration can be given due to issues experienced.
- We aspire to create novel research with startup potential, and we aspire to make a publishable research paper by the end of our FYP.
- When allocating work, we will assign workload difficulty (LOW to HIGH) to make sure the work amount is equally distributed. If we feel that a team member has worked more than others, this will be reflected in ITP Metrics.

ii. *Member Expectations*

- Team members are expected to commit a minimum of 12 hours per week in addition to two weekly 30 minute update meetings with supervisors.
- Team members are expected to attend the two 4 hour weekly workshop sessions.
- Team members are expected to communicate honestly and in good faith.

- Team members are expected to commit to delivering a high level of work, participate actively in meetings and provide constructive feedback.
- Team members are expected to abide by all university academic integrity policies as well as satisfy the creative commons licensing requirements.
- Team members are expected to be transparent in all use of generative AI.
- Any research conducted should be sourced or linked to avoid any plagiarism.

iii. Role Expectations

- For weekly meetings, the role of chairperson, minute taker and timekeeper will alternate in each meeting.
- Chairperson: Responsible for ensuring discussions are in accordance with the meeting agenda.
- Minute Taker: Responsible for taking accurate and reflective notes of meeting discussion points.
- Timekeeper: Responsible for ensuring meetings remain on track and on time during meetings and don't go on too big of a tangent.
- As the project further develops, more roles may be described to reflect the relevant duties of team members.

B. Communication

i. Communication Medium

- Team members are expected to communicate to our team members using Messenger.
- Team members are expected to communicate with the supervisors using email.
- We plan to have in person communication during the week (two weekly workshop sessions) as well as potentially incorporate zoom, if people cannot attend on campus due to a valid reason.
- Notion, Google Drive and GitHub are also expected to be used for sharing files, information, and tracking progress of the project.

ii. Communication Timelines

- Messages are acceptable anytime, though a response is only required within 5 hours on weekdays 9am – 9pm and 24 hours all other times.
- Calls should be within reasonable working hours (9am – 7pm, weekdays)
- Response times are expected to be quicker, within 3 hours, within two weeks of a submission deadline.

iii. Communication Code of Conduct

- Members are expected to remain professional while amicable.
- Members are expected to try to keep all personal questions not relevant to work outside of internal and external communications.
- Members are expected to provide constructive criticism in a respectful manner.

C. Team Meetings

i. Scheduling

- Members are expected to attend the two recurring meetings with supervisors on Mondays and Thursdays.
- Members are also expected to attend the two workshop sessions every week.
- If unanticipated time conflicts arise two times or more, then a new proposed meeting and workshop times need to be arranged.

ii. Involvement

- Any Zoom meetings require appropriate meeting etiquette, with include camera on and mic on (unless noisy background noise, which in that case would require the members to only turn their audio on when speaking).
- Everyone will be called on during the meetings to present at least one thing, and it is everyone's responsibility to be proactive and call on each during the meeting.
- Ensuring that roles such as chairperson, minute taker and timekeeper alternate for each meeting will result in fair involvement from all team members.
- All team members will be involved in the decision making process, using an anonymous poll for major decisions in the project, to ensure that fair decision making is made that everyone is aligned on.

iii. Attendance & Notice

- Members are expected be on time with an allowance of 5 minutes in normal circumstances and 10 minutes late with reasonable justification.
- Otherwise, if members are unable to attend meetings or running substantially late they are expected to communicate to the team via Messenger at their earliest with a reasonable justification and are expected to catch up via reading the meeting minutes and asking questions in messenger. There will be no make-up meetings, it is the team member's responsibility to catch up by reading the meeting minutes and clarify any questions they may have.

D. Team Conflict & Decision Making

i. Conflict Code of Conduct

- Members are expected to approach all conflict resolution with an open mind and willingness to hear and consider opposing viewpoints.
- Members should aim to communicate disagreement as early as possible to avoid issues building and remaining unaddressed.
- A full team meeting will be held, whereby different opinions can be shared, and anonymous voting system can be taken.
- Any interpersonal differences will be immediately escalated to a supervisor for further investigation.

ii. Initial Conflict & Conflict Escalation

- If conflict arises members will adhere to the following initial conflict negotiation steps:
 1. Each side should prepare with a list of what they think and why they think it (using professional and courteous language to support engineering arguments).
 2. We shall then schedule a whole team meeting to discuss the conflict.
 3. Before the meeting, the conflicting sides should share their points so other team members can read it in advance.
 4. Each team member shall present their thoughts within a fixed time.
 5. There will be a fixed time to allow for open discussion and rebuttals.
 6. We will then conduct an anonymous vote.
- If the above approach is not fruitful or there is severe disagreement about the verdict, the conflict shall be escalated to the manager in a whole team meeting.
 - o Keenan Granland will have the authority to settle final decisions for conflicts if they cannot be resolved by team members.
- If people are unable to be contacted within 2 days without a justifiable reason, this will be escalated to our manager.
 - o Keenan Granland will be a mediator to resolve this conflict.

iii. Decision-Making

- The meeting agenda should outline the decisions to be made in the session, allowing members to familiarise themselves with the matter in question beforehand.
- Members should provide all information relevant to the decision to be made in or prior to the meeting.
- Key impact decisions must be made collectively with unanimous agreement. If there are any disagreements, refer to the conflict resolution procedures detailed above.
- Team members responsible for task subsections may make minor decisions independently in accordance with team goals and interests.
- Keenan Granland will have the authority to settle final decisions for project tasks and goals if team members cannot agree through the conflict management procedure.

E. Stress Management

i. Monitoring & Assistance

- Members are expected to be open and communicate their weekly commitments in team meetings. This allows tasks to be redistributed and/or rescheduled. This activity is a part of the weekly meeting minutes template (In the meeting minutes, the chairperson will ask and record, how you are feeling/how stressed are you this week, how much workload do you have this week and how energetic do you feel).
- If any issues arise and members are concerned about other members help, members should contact a supervisor and/or Monash Connect to do an anonymous checkup .
- Members are encouraged to check in with other members and maintain a healthy working environment with a manageable workload for everyone.

ii. Resources

- [Monash Reducing Anxiety and Stress](#) including Monash Counselling and [Mental Health Resources](#).
- [Monash Counselling](#). (1300 788 336)
- [Monash Thrive](#) and [app](#).
- Monash Lunchtime mediation Tuesdays 1:15pm-1:45pm ([zoom link](#))
- Time management tools like Google Calendar.

F. Contract Code of Conduct

i. Contract Breaches

- A three-strike policy will be set in place. Each time the contract is breached that counts to a strike.
 - o A one-time violation of the contract without justification will result in a verbal warning in the next meeting.
 - o The second violation of the contract will result in a written email warning (with the manager, Keenan Granland, cc'ed).
 - o The third time violation of the contract will result in escalation with supervisor and a sit-down meeting.
- Breaches of team contract will be recorded in the weekly meeting minutes.

ii. Penalties

- If you receive three strikes (i.e. you committed three breaches of this contract), then this will be escalated to the manager (Keenan Granland), and this shall also be reflected in a poor ITP metric score.

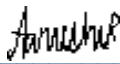
3. Declaration

By signing below, team members acknowledge and agree to be bound by the guidelines outlined above.



Team Member Signature

Date 29-Jul-2024



Team Member Signature

Date 29-Jul-2024



Team Member Signature

Date 29-Jul-2024



Manager Signature

Date 29-Jul-2024

Appendix D: Team Communication and Management Agreement

Team members have agreed to meet twice a week and have scheduled recurring meetings with their manager, as detailed in Table C.1. They have also established their preferred communication methods and the resources they will use for various processes, as outlined in Tables C.2 and C.3, respectively.

Table C.1: Table outlining the agreed up meeting times.

Verbal Communication	Timeslot Allocation
Team Standup Meetings (in-person, unless otherwise mentioned)	Mondays 1 pm-1:30 pm Thursdays 3:15 pm - 3:45 pm
Manager Meetings (with Keenan)	Weekly Thursdays 3:15 pm - 3:45 pm
Manager and Supervisor Meetings (with Chao and Keenan)	On Request
Team Workshop Sessions	Mondays 1pm to 4pm Thursdays 3pm to 6pm

Table C.2: Table outlining the preferred method of written communication.

Written Communication	Primary Median	Alternative Median
Among Team Members	Messenger Group Chat	Slack, LinkedIn, Email, Text, Call
With Keenan	Email	Text, Call
With Keenan and Chao	Email	Text, Call

Table C.3: Table outlining the different resources that will be used for decision processes.

Tool	Purpose
Google Drive	• Meeting Minutes • Research • Report Writing • File sharing (i.e. SolidWorks)
SolidWorks	• To develop CAD models and renders for the haptic glove
Unity	• To develop a VR environment that then can be connected to a VR headset and haptic devices
Git	• Code stored and shared
Notion	• Weekly Task Tracking to hold team members accountable to work and assign enough work fairly

Appendix E: Project Risk Assessment

62550	RISK DESCRIPTION	STATUS	TREND	CURRENT	RESIDUAL	
	ENG4701_CL_2024_S2_JasonAbiChebli_TahaSamani_AarushiRaheja_ImmersiveTeleroboticswithHapticFeedback	Live		Low	Not Assessed	
RISK TYPE						
4. Personal Risk Management Plan (Restricted Risk Assessment ONLY)						
RISK OWNER	RISK IDENTIFIED ON	LAST REVIEWED ON	NEXT SCHEDULED REVIEW			
Jason Abi Chebli	21/08/2024	29/08/2024	29/08/2027			
RISK FACTOR(S)	EXISTING CONTROL(S)	CURRENT	PROPOSED CONTROL(S)	TREATMENT OWNER	DUE DATE	RESIDUAL
Risks associated with Robotic Arm striking people.	Control: Robot speed is limited to 250mm/s in Testing Modes. Robot speed is limited to 500mm/s in Automatic Mode. Control Effectiveness: _____	Low				
	Control: Robot has torque limits of 20Nm on each joint. In the event of colliding with a person, the robot will lock. Control Effectiveness: _____					
	Control: Operators will be instructed to stand clear of the robot while it is in operation of new programs and to use T1 testing mode Control Effectiveness: _____					
	Control: Persons must only operate the robot if they have received training from KUKA or a person who has received training from KUKA is in the lab or close by. Control Effectiveness: _____					
Risks associated with pinch points of the robot/grippers and environment.	Control: Robot has torque limits of 20 Nm on each joint. In the event of colliding with a person, the robot will lock.	Low				

	Control Effectiveness: Control: Emergency stop button and safety mode on the pendant allow for immediate stop and adjustments of the robot. Operators will be instructed on how to use these. Control Effectiveness: Control: Operators will be instructed to stand clear of the robot while it is in operation of new programs and to use T1 testing mode. Control Effectiveness: Control: Persons must only operate the robot if they have received training from KUKA or a person who has received training from KUKA is in the lab or close by. Control Effectiveness:				
Risk associated with use of electrical devices. This includes but is not limited to using hand tools and basic power tools (e.g. power drills).	Control: All cables will be routed away from operator areas. Control Effectiveness: Control: I/O Pins will be deactivated if not in use. Control Effectiveness:	Low			
Risks associated with use of Low Voltage electricity and other prototyping activities such as soldering.	Control: Proper PPE to be worn. Safety glasses and gloves available in lab. Control Effectiveness: Control: Two-person rule. When attempting a potentially hazardous activity, one other lab member must be nearby and aware of the activity. Control Effectiveness:	Low			

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	Control: Only perform activities in prototyping or approved areas. Control Effectiveness: _____ Control: Follow safety guidelines of equipment, tools and materials. Control Effectiveness: _____				
Risks associated with tripping on cables/wires or other equipment on the floor	Control: Users are instructed to: 1. Conduct cable management to minimize loose cables and thus the chance of tripping. Control Effectiveness: _____	Low			
Risk associated with lab operations in a room with potential fire hazards.	Control: Fire extinguisher readily available within room. All persons within room are made aware of location during inductions. Control Effectiveness: _____	Low			
Risks associated with Ergonomics (i.e. looking at a laptop or monitor for long periods of time, having poor posture when working)	Control: Persons are instructed to follow Monash Office Ergonomics Guidelines. Control Effectiveness: _____	Negligible			
General hazard associated with using a VR headset	Control: Only perform task in an open space, clear from any objects and with minimal foot traffic. Control Effectiveness: _____ Control: Take breaks after periods of time to re-familiarise yourself with any changed surroundings Control Effectiveness: _____ Control: Use the buddy system to ensure someone else is keeping an eye out for you as you perform your VR tasks Control Effectiveness: _____	Low			

General hazards associated with using a haptic glove and full body haptic suit	Control: The haptic feedback is limited in design to prevent user discomfort by allowing users to override if needed Control Effectiveness: _____	Low				
	Control: Take breaks after periods of time to ensure that no longer-term strain is appearing. Control Effectiveness:					
General hazards associated with working after hours in Digital Twin Manufacturing Lab.	Control: Follow University policies regarding after hours work, e.g. buddy system, after hours procedures Control Effectiveness: _____	Low				
	Control: Minimise after hours work Control Effectiveness: _____					
	Control: Make sure to have enough breaks during the day and follow procedures in this risk assessment and relevant risk assessments of the given activity. Control Effectiveness: _____					
	Control: Minimise work after 9pm and never do so on two or more consecutive day. Make sure to have at least 6 hours of sleep before coming to work the next day Control Effectiveness:					
Slip, trip or fall while in the lab after hours in Digital Twin Manufacturing Lab.	Control: Inform your supervisor of intention to work after hours Control Effectiveness: _____	Low				
	Control: Have a Security number available on your mobile phone or contact					

	Security using the landline phone in the lab Control Effectiveness:				
Fatigue is possible if worked for long hours, after hours, in Digital Twin Manufacturing Lab.	Control: Take regular breaks Control Effectiveness:	Low			